Exploratory Data Analysis on the Automobiles Dataset

Introduction

The automobiles dataset contains information about 193 vehicles from different manufacturers. It includes vehicle characteristics such as body style, fuel type, engine size, horsepower, dimensions, fuel efficiency and price. The aim of this exploratory data analysis (EDA) was to clean the data, investigate patterns, and identify relationships between vehicle characteristics and pricing.

Description of the Dataset

After removing unnecessary columns, the dataset contained 193 observations and 24 variables. Key variables include make, fuel-type, body-style, drive-wheels, engine-size, horsepower, city-mpg, highway-mpg and price.

Data Cleaning

Several preprocessing and cleaning steps were performed:

- Removed the columns 'normalized-losses' and 'symboling' because they were not required for the analysis.
- Checked for duplicate records and confirmed that no duplicate rows existed.
- Replaced '?' values with missing values (NaN) and removed incomplete records.
- Converted numerical columns such as wheel-base, length, width, engine-size, bore, stroke, compression-ratio, horsepower, peak-rpm, city-mpg, highway-mpg and price to integer data types.
- Filtered and explored specific categories such as hatchback vehicles.

Missing Data

The dataset did not contain any missing values after the cleaning process. Potential missing values represented by '?' were replaced with NaN values and rows containing missing information were removed. After cleaning, the dataset contained 193 complete observations.

Data Stories and Visualisations

1. Most Expensive Cars

The five most expensive vehicles in the dataset were dominated by Mercedes-Benz, BMW and Porsche models. The most expensive car was a Mercedes-Benz priced at 45,400,

followed by BMW models priced above 40,000. This suggests that luxury manufacturers command significantly higher prices than mainstream brands.

2. Fuel Efficiency by Manufacturer

Average highway fuel consumption was compared across manufacturers using a bar chart. The analysis showed that some manufacturers produced considerably more fuel-efficient vehicles than others. Manufacturers with higher highway MPG values are more suitable for consumers seeking economical vehicles.

3. Engine Capacity Analysis

Vehicles with the largest engine capacities were primarily luxury brands such as Jaguar and Mercedes-Benz. These vehicles also tended to have high horsepower values and premium prices, indicating a positive relationship between engine size, performance and cost.

4. Manufacturer Representation

A count plot of vehicle manufacturers revealed that certain manufacturers contributed substantially more models to the dataset than others. This indicates that the dataset is not evenly distributed across manufacturers, with some brands being heavily represented.

5. Hatchback Vehicles

The dataset contained 63 hatchback vehicles. Hatchbacks represented a significant portion of the dataset, demonstrating their popularity across multiple manufacturers.

Conclusion

The exploratory analysis provided valuable insights into vehicle pricing, fuel efficiency and manufacturer trends. Luxury manufacturers generally produced vehicles with larger engines and higher prices, while fuel efficiency varied considerably between manufacturers. The cleaned dataset was suitable for further predictive modelling and statistical analysis.

This report was written by: Mmanoko Makibelo